SAFETY DATA SHEET

According to EC 2020/878 (REACH) and 1272/2008 (CLP)

Last update: September 12, 2024

1. IDENTIFICATION OF THE SUBSTANCE/ MIXTURE AND OF THE COMPANY/ UNDERTAKING

1.1 PRODUCT IDENTIFIER

Product name: Filament Noctuo PA12CF15

1.2 RELEVANT IDENTIFIED USES OF THE SUBSTANCE OR MIXTURE AND USES ADVISED AGAINST

Intended use: FDM 3D printing material

1.3 DETAILS OF THE SUPPLIER OF THE SAFETY DATA SHEET

Supplier:	**Manufacturer:**
Noctuo Sp. zo.o. ul. Sowińskiego 5 44-121 Gliwice, Polska	Noctuo Sp. zo.o. ul. Sowińskiego 5 44-121 Gliwice, Polska

1.4 EMERGENCY TELEPHONE NUMBER

Emergency telephone: +48 42 63 14 724

2. HAZARDS IDENTIFICATION

2.1 CLASSIFICATION OF THE SUBSTANCE OR MIXTURE

Classification according to Regulation (EC) No 1272/2008 of the European Parliament and of the Council (CLP):
This product is not classified as hazardous according to REGULATION (EC) No 1272/2008.

2.2 LABEL ELEMENTS

According to Council Regulation (EC) No 1272/2008 (CLP), the material is not classified as hazardous.

Warning word:	Not applicable
Hazard statement:	Not applicable
Precautionary statement:	Not applicable

2.3 OTHER HAZARDS

Fumes or vapors released during processing must not be inhaled. Molten material can cause burns. Dust and particles generated during use of the product can cause mechanical irritation of the eyes, skin and mucous membranes. Grinding of molded products can increase this phenomenon; therefore, inhalation of any dust in the environment should be avoided.

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1 SUBSTANCE

3.2 PREPARATION

Polyamide 12
Carbon fibers 15%

4. FIRST AID MEASURES

4.1 DESCRIPTION OF FIRST AID MEASURES

Following eye contact::	Rinse with plenty of water, also under the eyelids, for at least 15 minutes. Consult a doctor.
Following skin contact:	In case of burns caused by melted product, immediately cool the burned area, preferably under running water. DO NOT remove the material from the skin. Removal may cause serious tissue damage. Seek medical attention
Following inhalation:	Exposed to the inhalation of volatile compounds decomposition should be removed from the contaminated atmosphere, provide him with a supply of fresh air, protect from loss of heat; call a doctor.
Following ingestion:	Seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.

4.2 MOST IMPORTANT SYMPTOMS & EFFECTS BOTH ACUTE & DELAYED

Mechanical irritation caused by the reaction of product particles.

4.3 INDICATION OF ANY IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT NEEDED

Move the injured person to a ventilated room and consult a doctor.

5. FIRE - FIGHTING MEASURES

5.1 EXTINGUISHING MEDIA

Suitable extinguishing agents: All types of extinguishing agents (water, foam, carbon dioxide, extinguishing powder, etc.)

Unsuitable extinguishing agents: None.

5.2 SPECIAL HAZARDS ARISING FROM THE SUBSTANCE OR MIXTURE:

Decomposition products released during fire:
Carbon monoxide, carbon dioxide, nitrogen oxides (NOx), low molecular weight hydrocarbons, azo compounds. Under specific fire conditions, the presence of traces of other toxic substances cannot be ruled out. The formation of additional decomposition and oxidation products depends on the fire conditions.

5.3 ADVICE FOR FIREFIGHTERS

Wear self-contained breathing apparatus and fire-resistant clothing.

6. ACCIDENTAL RELEASE MEASURES

6.1 PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT & EMERGENCY PROCEDURES

Ensure adequate ventilation.

6.2 ENVIRONMENTAL PRECAUTIONS

Do not allow product to reach sewage system or water bodies.

6.3 METHODS AND MATERIAL FOR CONTAINMENT AND CLEANING UP

Do not dispose of into sewers or soil.

6.4 REFERENCE TO OTHER SECTIONS

For information on personal protection and waste disposal, see sections 8.2 and 13.

7. HANDLING AND STORAGE

7.1 PRECAUTIONS FOR SAFE HANDLING

Use only for the intended purpose (3D printing) on technically efficient devices, using the recommended parameters and in good ventilation conditions.

During handling, avoid the formation of significant quantities of particles with a granulometry of less than 500 micrometers, referring in these cases to the guidelines contained in the NFPA 654 (National Fire Protection Association) standard or similar.

Apply appropriate measures to prevent the formation of electrostatic discharges (grounding of equipment, etc.) in accordance with the guidelines of the CLC/TR 50404 Instruction (Electrostatics - Code of practice for the avoidance of hazards due to static electricity) or other similar guidelines.

The material contains carbon fibers; therefore, it is necessary to check the level of protection of the workplace, electrical devices and in particular compliance in the presence of conductive dust.
Do not touch the molten material.

7.2 CONDITIONS FOR SAFE STORAGE, INCLUDING ANY INCOMPATIBILITIES

Store closed in a dry place, protected from atmospheric conditions.

7.3 SPECIFIC END USE(S)

The product is designed for use as a filament for 3D printing using FDM technology.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 CONTROL PARAMETERS

Occupational Exposure Limits for Contaminants (ACGIH)

TLV (NDS):	10 mg/m3	TLV-TWA	Inhalable fumes
	3 mg/m3	TLV-TWA	Inhalated dust

DEFINITIONS
TLV- TWA (Time-average exposure limit): average concentration calculated over an 8-hour workday and a 40-hour workweek that does not cause harmful effects in exposed workers.

Derived No Effect Levels (DNEL): No data available.

Predicted No Effect Concentration in the environment (PNEC): No data available.

8.2 EXPOSURE CONTROL

During the use of the material and processing of elements made of it, provide appropriate means and protective devices to remove any dust in the work environment. During printing, remove fumes or vapors using an appropriate exhaust system. In the case of emission of pollutants into the atmosphere during the processing of plastic materials, observe their concentration limits established by the competent authorities and legally binding. Do not touch the molten material.

Eye protection:	When handling in the presence of dust, it is recommended to use safety glasses EN 166. When processing in the presence of liquid material, it is recommended to use a protective shield.
Skin and body protection:	When handling in the presence of dust, it is recommended to use gloves EN 388 (2132) and protective clothing. When processing in the presence of fumes and dust, it is recommended to use protective clothing and gloves marked EN 388 (4131), EN 407 (X2XXXX), EN 374-3.
Respiratory protection:	When using and processing the material in the presence of dust or gas/vapour, it is recommended to use an FFP2 protective mask.
Hand protection:	Gloves EN 388 (4131), EN 407 (X2XXXX), EN 374-3.
Other hazards:	Users should be protected from the possibility of contact with molten material during processing.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 INFORMATION ON BASIC PHYSICAL AND CHEMICAL PROPERTIES

Physical state	Solid
Appearance:	Circular profile
Odour:	Characteristic
pH:	Not applicable
Melting point:	270-290 °C
Vapour pressure:	Not applicable
Vapour density:	Not applicable
Density:	1.06 g/cm3
Flash point:	350 °C
Boiling temperature:	Not available
Inflammability	It is not flammable
Explosive properties:	It is not explosive in the form introduced for sale
Auto-ignition temperature:	400 °C

9.2 OTHER SAFETY INFORMATION:

Solubility in water: insoluble at 20°C.

10. STABILITY AND REACTIVITY

10.1 REACTIVITY

No hazardous reactions if requirements/directions regarding storage and handling are observed.

10.2 CHEMICAL STABILITY

Stable product if storage and handling requirements/instructions are followed.

10.3 POSSIBLE HAZARDOUS REACTION

No dangerous reactions known.

10.4 CONDITIONS TO AVOID

Before processing, it is recommended to dry the product according to the technical instructions. Note! If the material is used at a temperature higher than the highest suggested, a slight decomposition may occur; the degree of decomposition increases with longer residence time in the head. Avoid contamination with other materials that may create harmful gases and fumes during printing. During cleaning, do not allow fumes from the melted material to spread into the work environment. For additional information, refer to the recommendations in the product data sheet.

10.5 INCOMPATIBLE MATERIALS

Avoid contamination with other materials that could produce harmful gases and fumes during the transformation phase.

10.6 HAZARDOUS DECOMPOSITION PRODUCTS

Low molecular weight hydrocarbons, azo compounds.

11. TOXICOLOGICAL INFORMATION

11.1 INFORMATION ON TOXICOLOGICAL EFFECTS

No specific studies have been conducted to determine the degree of toxicity of the product. The assessment is based on information about similar products and individual ingredients and is based on professional experience and technical literature.

Corrosive effect: The product may cause mechanical skin irritation.

Mutagenic and carcinogenic effect: No data available.

Sensitizing effect on the respiratory tract: No data available.

Sensitizing effect on the skin: No data available.

Serious eye damage, irritation effect on the eyes: The product may cause mechanical eyes irritation.

11.2 INFORMATION ON OTHER HAZARDS

No data.

12. ECOLOGICAL INFORMATION

12.1 TOXICITY

Water treatment plants: The material may be removed from water by mechanical separation. In accordance with EU and national regulations, water in contact with the material or pressed/printed parts may require special treatment before discharge to the sewage system. If necessary, ensure, in accordance with EE and national regulations, treatment of exhaust gases from fume extraction plants during use of the material.

12.2 PERSISTENCE AND DEGRADABILITY

Potentially non-biodegradable. Expected to be durable.

12.3 BIOACCUMULATIVE POTENTIAL

It is not expected to bioaccumulate.

12.4 MOBILITY IN SOIL

Based on the morphology and composition of the product, it is not possible to penetrate the substrate.

12.5 RESULTS OF PBT AND VPVB ASSESSMENT

The material does not contain PBT (persistent, bioaccumulative, toxic) or vPvB (very persistent, very bioaccumulative) substances.

12.6 OTHER ADVERSE EFFECTS

There are no known significant effects and no critical hazards.

12.7 ADDITIONAL INFORMATION

Use the product in accordance with well-proven working methods, preventing its release into the environment.

13. DISPOSAL CONSIDERATION

13.1 WASTE TREATMENT METHODS

The material must be recycled, disposed of or incinerated in accordance with applicable local and national regulations. Anything that cannot be recycled or recovered must be taken to an appropriate facility.

Dispose of packaging and waste in accordance with applicable local and national regulations.

14. TRANSPORT INFORMATION

14.1 UN number

Not applicable.

14.2 UN proper shipping name

Not applicable.

14.3 Transport hazard class(es)

Not applicable.

14.4 Packing Group

Not applicable.

14.5 Environmental hazards

Not applicable.

14.6 Special precautions for user

Not applicable.

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable.

15. REGULATORY INFORMATION

15.1 SAFETY, HEALTH AND ENVIRONMENTAL REGULATIONS /LEGISLATION SPECIFIC FOR THE SUBSTANCE OR MIXTURE

Legal references

Classification and marking:
- Dir. 2001/60/EC - Dir. 1999/45/EC - Dir. 92/32/EC - Dir. 67/548/EEC and its amendments - Council (EC) No. 1272/2008.

Occupational health and safety:
- D.M. 26/02/2004 - D.Lgs. 233/03 "ATEX" - Dir. 98/24/EC, 89/391/EEC, 89/654/EEC, 2009/104/EC, 89/656/EEC, 2004/37/EC, 2000/54/EC, 2003/10/EC, 2009/148/EC - D.Lgs. 81/2008 - D.Lgs. n. 106 03/08/2009.

atmospheric emissions:
- D.Lgs. n. 152 03/04/2006 - DM 12/7/94 - Dir. 2008/50/EC - Dir. 2010/75/EU.

water protection:
- D.Lgs. n. 219 of 10/12/2010 - D.Lgs. n. 152 of 03/04/2006 - Dir. 91/271/EEC, 2000/60/EC, 2008/105/EC, 2009/90/EC, 2013/39/EU.

waste disposal:
- D.Lgs. n. 152 03/04/2006 - Dir. 2008/98/EC, 94/62/EC, 2001/118/EC.

WOO:
- D.Lgs. 475/92 - D.Lgs. 10/97 - D.M. 02/05/2001 - Dir. 89/686/EEC - Dir. 93/68/EEC - Dir. 93/95/EEC - Dir. 96/58/EC.

This technical data sheet has been prepared in accordance with the following standards: - Regulation (EU) No. 453/2010 - Regulation (EC) No. 1272/2008 - Regulation (EC) No. 1907/2006 (REACh) - Ministerial Decree 07/09/2002 - Dir. 2001/58/EC - Dir. 1999/45/EC - ISO 11014:2009.

15.2 CHEMICAL SAFETY ASSESSMENT

For this substance a chemical safety assessment is not yet required.

16. OTHER INFORMATION

The information contained in this sheet is based on current knowledge and experience with the product. This information is intended to support independent assessment of the methods used to ensure safe work with the filament and its safe disposal.